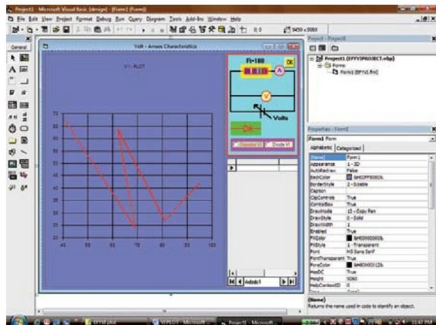




Screenshot 3: Visual Basic form

```
Adodc1.Recordset.MoveNext
```

```
Wend
```

```
DataGrid1.Refresh
```

```
MSChart1.Refresh
```

```
End Sub
```

In 'option' button-click event, the selection of either resistor or diode is made by setting their left and top properties. The V-I plot changes accordingly by setting 'RecordSource' property of Adodc1 for each table. The resistor frame becomes invisible when 'diode' option is selected:

```
Private Sub Plots_Click(Index As Integer)
```

```
Plot1 = Index
```

```
Select Case Index
```

```
Case 0:
```

```
ctop = Resistor.Top
```

```
cleft = Resistor.Left
```

```
Resistor.Top = Diode.Top
```

```
Resistor.Left = Diode.Left
```

```
Diode.Top = ctop
```

```
Diode.Left = cleft
```

```
Text1.Visible = True
```

```
Label3.Visible = True
```

```
OKButton.Visible = True
```

```
MSChart1.TitleText =
```

```
"Resistor VI Characteristics"
```

```
Adodc1.RecordSource =
```

```
"OHMSLAW"
```

```
Adodc1.Refresh
```

Case 1:

```
ctop = Diode.Top
```

```
cleft = Diode.Left
```

```
Diode.Top = Resistor.Top
```

```
Diode.Left = Resistor.Left
```

```
Resistor.Top = ctop
```

```
Resistor.Left = cleft
```

```
Text1.Visible = False
```

```
Label3.Visible = False
```

```
OKButton.Visible = False
```

```
MSChart1.TitleText =
```

```
"Diode VI Characteristics"
```

```
Adodc1.RecordSource =
```

```
"DIODEVI"
```

```
Adodc1.Refresh
```

End Select

```
Adodc1.Recordset.Sort =
```

```
"VOLTS ASC"
```

```
DataGrid1.Refresh
```

```
MSChart1.Refresh
```

End Sub

The datagrid controls either datagrid1_aftercoledit() event or datagrid1_keyup() event to calculate current for any change in voltage. The 'ok' button-click event calculates current for any change in resistor value at run time and also updates all entries in the table.

EFY note. The source code of this program is included in this month's EFY-CD.

Both the authors are lecturers at Government Polytechnic, Bidar, Karnataka

EFYVIFRM

Option Explicit

Dim V, I, R As Single

Dim rtop, rleft, dtop, dleft, ctop,

cleft, PLOT1 As Integer

Private Sub Form_Load()

On Error Resume Next

R = Val(Text1.Text)

Plots(1).Value = True

Plots(1).Value = False

Plots(0).Value = True

MSChart1.Plot.UniformAxis = False

MSChart1.Plot.Axis(VtChAxisIDx).AxisTitle = "Voltage"

MSChart1.Plot.Axis(VtChAxisIDy).AxisTitle = "Current"

MSChart1.Plot.Axis(VtChAxisIDy).AxisTitleLayout.Orientation = VtOrientationVertical

DIODE_Adoajust

OHMS_Adoajust

End Sub

Private Sub OKButton_Click()

R = Val(Text1.Text)

OHMS_Adoajust

End Sub

Private Sub Plots_Click(Index As Integer)

PLOT1 = Index

Select Case Index

Case 0:

```
ctop = Resistor.Top
```

```
cleft = Resistor.Left
```

```
Resistor.Top = Diode.Top
```

```
Resistor.Left = Diode.Left
```

```
Diode.Top = ctop
```

```
Diode.Left = cleft
```

```
Text1.Visible = True
```

```
Label3.Visible = True
```

```
OKButton.Visible = True
```

MSChart1.TitleText = "Resistor VI

Characteristic"

Adodc1.RecordSource = "OHMSLAW"

Adodc1.Refresh

Case 1:

```
ctop = Diode.Top
```

```
cleft = Diode.Left
```

```
Diode.Top = Resistor.Top
```

```
Diode.Left = Resistor.Left
```

```
Resistor.Top = ctop
```

```
Resistor.Left = cleft
```

```
Text1.Visible = False
```

```
Label3.Visible = False
```

```
OKButton.Visible = False
```

```
MSChart1.TitleText = "Diode VI
```

Characteristic"

Adodc1.RecordSource = "DIODEVI"

Adodc1.Refresh

End Select